

Git Branch Works Structure (Full + Diagram)

1. What is a Branch in Git?

A branch is a pointer (reference) to a commit. It does not copy files.

2. What is a Commit?

A commit is a snapshot of your project. Each commit has a unique ID and a parent commit.

3. What is a Parent Commit?

Each commit (except the first one) points to a previous commit.

Example: C1 → C2 → C3

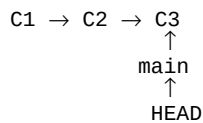
4. What is main Branch?

main is the default branch pointing to the latest commit.

5. What is HEAD?

HEAD is a pointer to the current working branch.

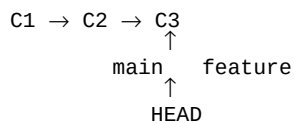
6. Initial State Diagram



7. Creating a New Branch

Command: `git branch feature`

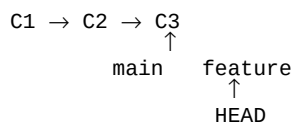
New branch points to same commit. HEAD does not move.



8. Switching Branch

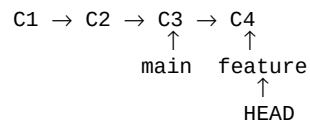
Command: `git switch feature`

HEAD now moves to feature branch.



9. New Commit in Feature Branch

After commit:



Explanation:

- C4 parent is C3
- Only feature branch moves
- main remains unchanged

10. Key Concepts

- Branch = pointer
- HEAD = current branch
- Commit = snapshot
- Parent = previous commit

11. Final Summary

- main points to latest commit
- HEAD shows current branch
- New branch starts from same commit
- New commits move only active branch